canon-0.8.0-research-track

Companion research-track sectors migrated from the main SST canon

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Abstract

This companion document collects those sectors that remain useful for continuity and future development but are too long, too provisional, or too phenomenology-specific for the core body of *Swirl-String-Theory_ Canon-v0.8.0.* Their migration out of the main canon is editorial rather than negative: each subsection remains part of the broader SST research program, but none is required for the minimal logical closure of the core canon.

1 Research-track sectors migrated from the main canon

1.1 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector. Let

$$T_{\mathrm{eff}} = T_{\mathrm{eff}}\big(\rho_E,\, \omega,\, S^{\circlearrowleft}_{(t)}\big) \tag{1}$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff}.

[ORTHODOX]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[SPECULATIVE]: SST does not yet derive the constitutive relation above, so this sector remains a research program rather than canonically closed physics.

1.2 Minimal Coupling Interface to QED

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \to \nabla - i\frac{m}{\hbar}A_{\mathrm{eff}}, \qquad A_{\mathrm{eff}} = \chi\, v, \tag{2}$$

where χ is a scalar weight and v is the local carrier velocity field.

[ORTHODOX]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[SPECULATIVE]: the above substitution is only an analogy template unless a full gauge-covariant derivation is supplied.

1.3 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Calugareanu–White–Fuller relation

$$Lk = Tw + Wr \tag{3}$$

as a refinement of the knot-based matter/antimatter dictionary.

[ORTHODOX]: this is a standard geometric identity for framed curves and ribbons.

[DERIVED]: it can be used internally to separate writhe-dominated from twist-dominated realizations of the same coarse knot class.

[SPECULATIVE]: the mapping from $\mathrm{sign}(Tw)$ or related chirality measures to particle/antiparticle sectors remains a model hypothesis.

1.4 Particle-Candidate Mapping Layer

[SPECULATIVE] Working dictionary: a minimal knot/composition mapping retained from earlier SST work is summarized in Table 1. This table is organizational rather than definitive; it records the current topological assignments used by the mass and taxonomy program.

Knot class / composition	Candidate	Status / role
3_1	electron	baseline lepton calibration
$T(2,5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2,7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
$5_2 + 5_2 + 6_1$	proton	composite baryon candidate
analogous nearby composite class	neutron	composite baryon candidate

Table 1: Minimal knot-class to particle-candidate summary retained as a research-track dictionary.

[CRITICAL NOTE]: Table 1 is a working dictionary only. A full canonical assignment requires an explicit invariant set and a mass-functional comparison program.

1.5 Hydrodynamic Exchange and the Pauli Barrier

[ORTHODOX]: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\ell_1 \cdot d\ell_2}{\|r_1 - r_2\|}. \tag{4}$$

[DERIVED]: if two identical electron-candidate loops are forced into the same spatial state, regularization at the core radius r_c yields the overlap penalty

$$V_{\text{Pauli}}(d) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \iint \frac{d\ell_1 \cdot d\ell_2}{\sqrt{\|r_1(\theta_1) - r_2(\theta_2) + d\|^2 + {r_c}^2}}, \tag{5}$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\max} \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{r_c}\right) \mathcal{O}(1). \tag{6}$$

Using the canonical values

$$\rho_f = 7.0 \times 10^{-7}\ \mathrm{kg\,m^{-3}}, \tag{7}$$

$$r_c = 1.40897017 \times 10^{-15}\ \mathrm{m}, \tag{8}$$

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\| \approx 9.6836 \times 10^{-9}\ \mathrm{m^2\,s^{-1}}, \tag{9}$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, one obtains the benchmark estimate

$$V_{\text{Pauli}}^{\max} \sim 1.23 \times 10^{-18}\ \mathrm{J} \sim 7.6\ \mathrm{eV}. \tag{10}$$

[CRITICAL NOTE]: this is retained as a canonical scale benchmark, not yet as a closed theorem. Its role is to show that the exclusion-energy estimate falls in an atomically relevant window rather than at obviously pathological scales.

1.6 Numerical Benchmark Layer

[DERIVED] Computational-program requirement: earlier SST work contained explicit benchmark tables for particle and atomic masses. In v0.8.0, the benchmark layer is retained as a reproducibility requirement rather than as a stand-alone authority claim.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode class
e	0.51099895	0.51099895	0	predictive/closure
μ	105.6583745	...	...	predictive candidate
τ	1776.86	...	...	predictive candidate
p	938.272088	...	...	predictive or closure
n	939.565420	...	...	predictive or closure

Table 2: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

[CRITICAL NOTE]: a future v0.8.x benchmark appendix should include the exact script or package path used for generation, archived CSV outputs, mode definitions (predictive vs. exact-closure), and uncertainty propagation on canonical constants and geometric factors.

1.7 Gauge-Sector Roadmap

[SPECULATIVE] Structural roadmap: a retained minimal statement of the gauge program is

$$g_{\text{eff}} = su(3) \oplus su(2) \oplus u(1), \tag{11}$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A^a_\mu(x) = \partial_\mu \theta^a(x), \tag{12}$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A^a_\mu \mapsto A^a_\mu + q^a \partial_\mu \chi. \tag{13}$$

[CRITICAL NOTE]: the gauge layer is retained only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.